

»» Chapter VI ««

LAURENT EXPANSION. MEROMORPHIC FUNCTIONS

PREFATORY REMARKS

In this chapter we first consider functions analytic in an annulus. At this stage of development there are no surprises in store. The way has been paved by the Cauchy theory; the Cauchy formula for an annulus lies at hand and because of the special character of an annulus the resulting expansion theorem is in the nature of things. In simple cases it is possible to determine explicitly the expansions whose existence is assured. When we concentrate on the case of an annulus where one of the radii is either 0 or $+\infty$, we find that we can obtain an important correlation between the behavior of the function near the center of the annulus (or near ∞ , as the case may be) and that of the sequence of the coefficients of the expansion. We are concerned here with the *isolated singularities* of an analytic function. Earlier we met the Riemann condition (Ex. 3.10, Ch. V) which guarantees analyticity at the point in question for a suitable extension of the function. The methods of the present chapter give another approach to the problem. We shall see that two possibilities other than that referred to arise in the unrestricted study of isolated singularities of analytic functions. One is that the function has the limit ∞ at the point in question, the other that the function does not possess a limit (in $\hat{\mathbb{C}}$) at the point in question. In this final case the function has the interesting characterizing property that the image of every neighborhood of the point less the point lying in the domain of the function is dense in $\hat{\mathbb{C}}$, that is, has $\hat{\mathbb{C}}$

as its closure (theorem of Weierstrass). The celebrated “big Picard theorem” (cf. Ch. XVI) gives a far sharper statement.

We are led very naturally to study continuous maps f of O open ($\subset \hat{\mathbb{C}}$) into $\hat{\mathbb{C}}$ which are analytic at each point of O save at the points of a set clustering at no point of O . We shall see that we shall be able to give a reasonable definition of “analytic at ∞ ,” so that if $\infty \in O$ it will turn out that, when $f(\infty) \neq \infty$, f is analytic at ∞ . The functions that we are describing are termed *meromorphic on O* (“fraction” or more accurately “part” shaped). Later (Ch. VIII) we shall see that, when $O \neq \hat{\mathbb{C}}$, the functions meromorphic on O are simply the quotients of functions analytic on O , the appropriate definitions prevailing at the points where the denominators take the value zero. The *rational functions* are precisely the functions meromorphic on $\hat{\mathbb{C}}$, as we shall see (Ex. 4.21, this chapter). They enjoy many properties interesting from a function-theoretic point of view.†

1. LAURENT EXPANSION

Let us turn at once to the basic representation theorem.

Theorem 1.1: *Let $a \in \mathbb{C}$ and let $0 \leq r_1 < r_2 \leq +\infty$. Let $\mathbf{A}$ denote the family of functions analytic on the annulus $A = \{z: r_1 < |z - a| < r_2\}$. Let $\mathbf{B}$ denote the family of sequences $(b_k)_{k \in \mathbb{Z}}$ such that $\sum_0^\infty b_k z^k$ converges when $|z| < r_2$ and $\sum_1^\infty b_{-k} z^k$ converges when $|z| < 1/r_1$. Then there exists a unique map $\mathbf{L}: \mathbf{A} \rightarrow \mathbf{B}$ such that if $(b_k) = \mathbf{L}(f)$, then*

$$f(z) = \sum_0^\infty b_k (z - a)^k + \sum_1^\infty b_{-k} (z - a)^{-k}, \quad z \in A. \quad (1.1)$$

The map $\mathbf{L}$ is univalent, onto, and linear.

PROOF: The unicity is easy to establish. For starting with (1.1) we have

$$b_k = \frac{1}{2\pi i} \oint_{C(a; r)} f(z) (z - a)^{-(k+1)} dz, \quad k \in \mathbb{Z}, \quad (1.2)$$

where $r_1 < r < r_2$, as follows readily on term-by-term integration. Now let $f \in \mathbf{A}$, and suppose that $r_1 < \rho_1 < \rho_2 < r_2$. By the Cauchy integral formula for

† A rational function is a function with domain $\hat{\mathbb{C}}$ associated with an ordered pair of polynomials (P, Q) , Q not the constant zero, by the requirements that its value at $z \in \mathbb{C}$, $Q(z) \neq 0$, is $P(z)/Q(z)$ and that it is continuous on $\hat{\mathbb{C}}$. Exactly one such function is associated with an allowed (P, Q) .

an annulus (Ex. 3.6, Ch. V) we have

$$f(z) = \frac{1}{2\pi i} \oint_{C(a; \rho_2)} \frac{f(w)}{w - z} dw + \frac{1}{2\pi i} \oint_{C(a; \rho_1)} \frac{f(w)}{z - w} dw,$$

$\rho_1 < |z - a| < \rho_2$. By Ex. 3.10, Ch. V we know that

$$\frac{1}{2\pi i} \oint_{C(a; \rho_2)} \frac{f(w)}{w - z} dw = \sum_0^{\infty} b_k (z - a)^k, \quad |z - a| < \rho_2,$$

where b_k is given by (1.2) with $r = \rho_2$. By the cited result we also have

$$\frac{1}{2\pi i} \oint_{C(a; \rho_1)} \frac{f(w)}{z - w} dw = \sum_1^{\infty} b_{-k} (z - a)^{-k}, \quad |z - a| > \rho_1,$$

where b_k is given by (1.2) with $r = \rho_1$. By the Cauchy integral theorem for an annulus for given $f \in \mathbf{A}$ the integrals on the right side of (1.2) are independent of admitted r . Hence for $f \in \mathbf{A}$ with b_k given by the integrals (1.2) we have (1.1) as well as the convergence of $\sum_0^{\infty} b_k z^k$ and $\sum_1^{\infty} b_{-k} z^{-k}$ for the stated values of z . The map $\mathbf{L}$ has been thereby exhibited.

That $\mathbf{L}$ is univalent is a trivial consequence of (1.1). Its linearity follows from the linearity of the integral. To show that $\mathbf{L}(\mathbf{A}) = \mathbf{B}$, we consider $(b_k) \in \mathbf{B}$ and note that the right side of (1.1) defines a function f analytic on A . Clearly $\mathbf{L}(f) = (b_k)$.

We recall that an annulus $A = \{r_1 < |z - a| < r_2\}$ determines (a, r_1, r_2) (cf. Ex. 6.8, Ch. III). Hence given an annulus A exactly one a and one $\mathbf{L}$ come into consideration, there being exactly one parameter triple (a, r_1, r_2) specifying A . Given $f \in \mathbf{A}$, let $(b_k) = \mathbf{L}(f)$. We term b_k the k th *Laurent coefficient* of f . We term the identity (1.1) the *Laurent representation* or *expansion* of f . More generally, if f is analytic in A , by the *Laurent coefficients* and *Laurent expansion of f relative to A* are meant, respectively, the Laurent coefficients and Laurent expansion of $f|A$.†

Given $(b_k): \mathbf{Z} \rightarrow \mathbf{C}$, it is reasonable to ask for I , the intersection of the convergence sets of $\sum_0^{\infty} b_k z^k$ and $\sum_1^{\infty} b_{-k} z^{-k}$. Let r_2 be the radius of convergence of $\sum_0^{\infty} b_k z^k$ and let r_1 denote the radius of convergence of $\sum_1^{\infty} b_{-k} z^{-k}$. Then $I \subset \{z: 1/r_1 \leq |z| \leq r_2\}$. If $I \neq \emptyset$, then $1/r_1 \leq r_2$. We term

$$\left(\sum_0^{\infty} b_k z^k, \sum_1^{\infty} b_{-k} z^{-k} \right)$$

the *Laurent pair associated with (b_k)* . The function assigning to $z \in I$ the value $\sum_0^{\infty} b_k z^k + \sum_1^{\infty} b_{-k} z^{-k}$ is termed the *Laurent sum associated with (b_k)* . When $1/r_1 < r_2$, the Laurent sum associated with (b_k) is analytic in $\{1/r_1 < |z| < r_2\}$.

† It is banal that the notions are concordant when the domain of f is A .

We shall use the notation $\sum_{-\infty}^{+\infty} b_k z^k$ to denote the Laurent pair and the Laurent sum associated with (b_k) . Ambiguity will be eliminated by context.

In general, $\sum_{-\infty}^{+\infty} a_k$ unglossed will refer to the situation where $\sum_0^{\infty} a_k$ and $\sum_1^{\infty} a_{-k}$ are both convergent and will denote the sum $\sum_0^{\infty} a_k + \sum_1^{\infty} a_{-k}$.

It is worth our while from the point of view of technical facility to consider examples involving Laurent expansions. It will soon be apparent that functions represented additively as sums of functions having a "simple" character involve no great complications while functions represented multiplicatively as products may lead to computations that are cumbersome if not forbidding. The following example is instructive.

Let

$$f(z) = [(z-1)(z-2)]^{-1}, \quad z \in \mathbb{C} - \{1, 2\}.$$

Clearly f is analytic on the specified domain. We consider the annuli $A(a; r_1, r_2) = \{r_1 < |z-a| < r_2\}$ for the following sets of parameter triples (a, r_1, r_2) : $(0, 0, 1)$, $(0, 1, 2)$, $(0, 2, +\infty)$, $(1, 0, 1)$, $(2, 0, 1)$. The multiplicative approach is clumsy. In contrast, using the *partial fraction representation*, which we shall treat in a general setting in §2, this chapter—it has an additive character—we have a simple access to the calculation of the Laurent coefficients of $f|A(a; r_1, r_2)$ in the various cases. From the identity

$$f(z) = \frac{1}{z-2} + \frac{(-1)}{z-1},$$

the identity $z = (z-c) + c$, and the geometric series, we are led rapidly to the desired results. Thus

$$(z-2)^{-1} = \left[(-2)\left(1 - \frac{z}{2}\right)\right]^{-1} = \sum_0^{\infty} (-2^{-k-1})z^k, \quad |z| < 2;$$

$$(z-2)^{-1} = z^{-1}\left(1 - \frac{2}{z}\right)^{-1} = \sum_1^{\infty} 2^{k-1}z^{-k}, \quad |z| > 2;$$

$$(z-2)^{-1} = [(z-1) - (2-1)]^{-1} = \sum_0^{\infty} (-1)(z-1)^k, \quad |z-1| < 1;$$

$$-(z-1)^{-1} = \sum_0^{\infty} (1)z^k, \quad |z| < 1;$$

$$-(z-1)^{-1} = -z^{-1}\left(1 - \frac{1}{z}\right)^{-1} = \sum_1^{\infty} (-1)z^{-k}, \quad |z| > 1;$$

$$-(z-1)^{-1} = [(1-2) - (z-2)]^{-1} = \sum_0^{\infty} (-1)^{k+1}(z-2)^k, \quad |z-2| < 1.$$

It is now easy to determine the Laurent coefficients with the aid of Theorem 1.1. Thus for the first case we have

$$f(z) = \sum_0^{\infty} (1 - 2^{-k-1})z^k + \sum_1^{\infty} (0)z^{-k}, \quad 0 < |z| < 1.$$

Hence $b_k = 1 - 2^{-k-1}$, $k \geq 0$, and $b_k = 0$, $k < 0$. Attention is to be paid to the evasive "obvious" representation of 0 by $\sum_1^{\infty} (0)z^{-k}$. The same remark applies to

$$(z-1)^{-1} = \sum_1^{\infty} c_{-k}(z-1)^{-k}$$

with $c_{-1} = 1$, $c_{-k} = 0$, $k > 1$, etc. We may now determine from the above expansions the Laurent coefficients in the remaining cases. Thus for $(0, 1, 2)$ we have $b_k = -2^{-k-1}$, $k \geq 0$, and $b_k = -1$, $k < 0$; for $(0, 2, +\infty)$ we have $b_k = 0$, $k \geq 0$, and $b_k = 2^{-k-1} - 1$, $k < 0$; for $(1, 0, 1)$ we have $b_k = -1$, $k \geq -1$, and $b_k = 0$, $k < -1$; for $(2, 0, 1)$ we have $b_k = (-1)^{k+1}$, $k \geq -1$, and $b_k = 0$, $k < -1$.

In the exercises that follow we consider further properties of Laurent expansions, computational questions, and some applications.

EXERCISES

1.1. Treat the case $(0, 1, 2)$ of the above example by noting

$$[(z-1)(z-2)]^{-1} = \left[\sum_1^{\infty} z^{-k} \right] \left[\sum_0^{\infty} (-2^{-k-1})z^k \right], \quad 1 < |z| < 2,$$

and multiplying and gathering terms, justifying the calculation.

1.2. State and establish the paraphrase of Theorem 1.1 when A is replaced by $\Delta(a; r)$. Show that if f is analytic on $\Delta(a; r)$, then the k th Laurent coefficient of $f|[\Delta(a; r) - \{a\}]$ is $f^{(k)}(a)/k!$, $k \geq 0$, and is 0, $k < 0$.

1.3. Let f be analytic on A of Theorem 1.1. Let $r_1 < r < r_2$. Let I_r denote the analytic function on $\Delta(a; r)$ given by

$$\frac{1}{2\pi i} \oint_{C(a; r)} \frac{f(w)}{w-z} dw,$$

and let E_r denote the analytic function on $A(a; r, +\infty)$ given by

$$\frac{1}{2\pi i} \oint_{C(a; r)} \frac{f(w)}{z-w} dw.$$

Let $I = \bigcup_{r_1 < r < r_2} I_r$ and let $E = \bigcup_{r_1 < r < r_2} E_r$. Show that I is analytic on $\Delta(a; r_2)$,

that E is analytic on $A(a; r_1, +\infty)$ and $\lim_{\infty} E = 0$. Show further that $f(z) = I(z) + E(z)$, $z \in A$, and that f admits a unique representation of this form where I and E fulfill the requirements stated in the previous sentence. The demonstration is to be based on the Cauchy theorem and Cauchy formula for an annulus without appeal to the Laurent expansion.

1.4. Continuing with A, f, r as in Ex. 1.3, show that

$$\int_0^{2\pi} |f(a + re^{i\theta})|^2 d\theta = 2\pi \sum_{-\infty}^{+\infty} |b_k|^2 r^{2k}$$

where b_k is the k th Laurent coefficient of f (Parseval equality) and conclude that

$$|b_k| r^k \leq M(r), \quad k \in \mathbb{Z},$$

where

$$M(r) = \max\{|f(z)| : z \in C(a; r)\}$$

(Cauchy inequalities). Show that if equality holds for $k = j$, then $f(z) = b_j z^j$.

1.5. Continuing with f as above, show that $(k+1)b_{k+1}$ is the k th Laurent coefficient of f' . Note that the (-1) st Laurent coefficient of f' is zero. Show that a necessary and sufficient condition that f admit a primitive is that $b_{-1} = 0$.

1.6. Given A as above, let both f and g be analytic on A , let b_k denote the k th Laurent coefficient of f and c_k the k th Laurent coefficient of g . Let $h = fg$ and let d_k denote the k th Laurent coefficient of h . Prove that, for each $k \in \mathbb{Z}$, $\sum_0^\infty |b_j c_{k-j}|$ and $\sum_1^\infty |b_{-j} c_{k+j}|$ converge and $d_k = \sum_{-\infty}^{+\infty} b_j c_{k-j}$. Hint: Approximate $f(z)$ by

$$s_n(z) = \sum_{-n}^n b_k (z-a)^k$$

and calculate the Laurent coefficients of $s_n g$ as integrals.

Let $\alpha \in \mathbb{C}$. Apply the above result to determine the Laurent coefficients $b_k(\alpha)$ of

$$f(z) = \exp \left[\frac{\alpha(z - z^{-1})}{2} \right], \quad z \in \mathbb{C} - \{0\}.$$

Show by the use of the integral formulas for the Laurent coefficients that

$$b_k(\alpha) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \cos(\alpha \sin \theta - k\theta) d\theta.$$

Show that, when $k \geq 0$, the function b_k is the unique function w analytic on $\mathbb{C}$ satisfying the differential equation

$$z^2 w''(z) + z w'(z) + (z^2 - k^2) w(z) = 0,$$

and the conditions $w(0; w) = k$, $w^{(k)}(0) = 2^{-k}$. The function in question is the *Bessel function* of order k , customarily denoted J_k . Hint: Use power series methods on the expansion obtained for $b_k(\alpha)$.

1.7. Determine the Laurent expansions of the analytic square roots of $z^2 - 1$ with domain $A(0; 1, +\infty)$. Show that $(z-1)(z-2)$ with domain $A(0; 2, +\infty)$ has

analytic square roots and determine their expansions. Does $(z-1)(z-2)$ with domain $A(0; 1, 2)$ have an analytic square root?

1.8. Let f and g be analytic at a . We suppose that $g(a) = 0$, $n(a; g) < +\infty$. Let h be analytic at a and satisfy

$$g(z) = (z-a)^{n(a; g)} h(z),$$

z near a . Show that the k th Laurent coefficient of $f(z)/g(z)$ with domain $A(a; 0, r)$, r small, is the $[k + n(a; g)]$ th Laurent coefficient of $f(z)/h(z)$ with the domain $A(a; 0, r)$, r small. What assures that the coefficients in question are independent of admitted r ? What do we infer from the analyticity of $f(z)/h(z)$ at a ?

1.9. Let f be analytic at each point of $C(0; 1)$. Suppose that $f|C(0; 1)$ is univalent and that $f'(z) \neq 0$, $|z| = 1$. Show that for some annulus $A = A(0; r_1, r_2)$, $r_1 < 1 < r_2$, f is analytic in A and $f|A$ is univalent. Let g denote the inverse of $f|A$. Show that if f maps $C(0; 1)$ into itself, the k th Laurent coefficient of g relative to an annulus $A(0; \rho_1, \rho_2)$ contained in $f(A)$ where $\rho_1 < 1 < \rho_2$ is given by

$$\frac{O(0; \gamma)}{2\pi i} \oint_{C(0; 1)} \frac{z f'(z)}{[f(z)]^{k+1}} dz,$$

where $\gamma(t) = f(e^{2\pi i t})$, $t \in \mathbf{R}$.

Show that $f(z) = (3z + z^3)/(1 + 3z^2)$ is analytic at each point of $C(0; 1)$, that $f|C(0; 1)$ is a univalent map of $C(0; 1)$ onto itself, but that $f'(1) = 0$.

1.10. Let $0 < \varepsilon < 1$. Let $F(t) = t - \varepsilon \sin t$, with domain $\mathbf{R}$. Show that $F'(t) > 0$,

$$F(t + 2\pi) = F(t) + 2\pi, \quad t \in \mathbf{R}.$$

In particular, F is an increasing map of $\mathbf{R}$ onto itself. Let G denote the inverse of F . Show that

$$f = \{(e^{it}, e^{t(z - \varepsilon \sin z)}); z \in \mathbf{C}\}$$

fulfills the conditions of the preceding exercise. Show using the identity $g(e^{it}) = \exp[i G(t)]$, $t \in \mathbf{R}$, that the k th complex Fourier coefficient of G' is $J_{|k|}(|k|\varepsilon)$ and conclude that

$$G(t) = t + 2 \sum_1^\infty J_k(k\varepsilon) k^{-1} \sin kt.$$

The result of this exercise is of interest in the study of the Kepler equation encountered in celestial mechanics.

1.11. Let f be analytic on $A = A(a; r_1, r_2)$ and not take the value zero. Let $\Gamma_r(t) = a + re^{2\pi i t}$, $t \in \mathbf{R}$. Here $r_1 < r < r_2$. Let ν denote the common value of $O(0; f \circ \Gamma_r)$. Let $g(z) = f(z)(z-a)^{-\nu}$, $z \in A$. Show that the Laurent coefficient with index -1 of f'/f is ν and that g possesses an analytic logarithm. Show that f possesses an analytic logarithm if and only if $\nu = 0$ and that f has an analytic n th root, n a positive integer, if and only if n divides ν . Consider the special case where $f(z) = \prod_1^m (z - a_k)^{\nu_k}$, $a_k \in \mathbf{C} - A$, $\nu_k \in \mathbf{Z}$, $k = 1, \dots, m$.

Show that f admits a representation of the form $f(z) = I(z) E(z)$ where I is analytic on $\Delta(a; r_2)$ and $I(z) \neq 0$, E is analytic on $A(a; r_1, +\infty)$, E does not take the value 0, and $\lim_{\infty} E(z)(z-a)^{-\nu} = 1$. To what extent are I and E so specified unique?

[Such multiplicative decompositions were studied by G. D. Birkhoff for the

much more complicated situation of matrix-valued analytic functions [15, Vol. 1, 240–251].]

1.12. Let there be given a disk $\Delta(a; r)$, points $a_1, \dots, a_n \in \Delta(a; r)$, $r_1, \dots, r_n \in \mathbf{R}^+$ such that the sets $D_k = \{|z - a_k| \leq r_k\}$ satisfy $\bigcup_{1 \leq k \leq n} D_k \subset \Delta(a; r)$ and $D_j \cap D_k = \emptyset$, $j \neq k$. Let f be analytic on $D = \Delta(a; r) - \bigcup_{1 \leq k \leq n} D_k$. Let $s_k > r_k$ be such that $\{r_k < |z - a_k| \leq s_k\} \subset D$. Let E_k denote the function E and I_k the function I of Ex. 1.3, this set, relative to $f|_{\{r_k < |z - a_k| < s_k\}}$. Show that f admits a representation of the form

$$f(z) = g(z) + \sum_1^n E_k(z), \quad (1.3)$$

where g is analytic on $\Delta(a; r)$. Hint: Consider

$$g(z) = f(z) - \sum_1^n E_k(z), \quad z \in D,$$

$$g(z) = I_j(z) - \sum_{k \neq j} E_k(z),$$

$z \in \Delta(a_j; s_j)$, $j = 1, \dots, n$. Show that a representation of the type $g(z) + \sum_1^n E_k(z)$, where g is analytic on $\Delta(a; r)$ and E_k is analytic on $\mathbf{C} - D_k$, $\lim_{\infty} E_k = 0$, $k = 1, \dots, n$, is unique. Conclude from (1.3) that if $s < r$ satisfies $|a_k - a| + s_k < s$, $k = 1, \dots, n$, then

$$(a) \oint_{C(a; s)} f(z) dz = \sum_1^n \oint_{C(a_k; s_k)} f(z) dz,$$

and

$$(b) f(z) = \frac{1}{2\pi i} \oint_{C(a; s)} f(w) (w - z)^{-1} dw \\ - \sum_1^n \frac{1}{2\pi i} \oint_{C(a_k; s_k)} f(w) (w - z)^{-1} dw, \quad z \in \Delta(a; s) - \bigcup_{1 \leq k \leq n} \overline{\Delta(a_k; s_k)}.$$

These two results comprise of course the Cauchy integral theorem and integral formula for a closed disk less the union of a finite number of disks whose closures lie in the interior of the given disk and are mutually disjoint.

1.13. Let $f(z) = (\sum_0^n a_k z^k)/Q(z)$ where $Q(z) = z^{n+1} + \sum_0^n b_k z^k$, $n \in \mathbf{N}$, with domain $\mathbf{C} - \{Q(z) = 0\}$. Apply (1.3) of the preceding exercise to show that f admits a representation of the form

$$\sum_1^m P_k \left(\frac{1}{z - c_k} \right) \quad (1.4)$$

where $c_1, \dots, c_m$ are the distinct roots of $Q(z) = 0$ and P_k is a polynomial of degree $\leq n(c_k; Q)$ satisfying $P_k(0) = 0$. [Recall that the polynomial that is identically zero has degree $-\infty$. Generally, if $P(z) = \sum_0^\infty c_k z^k$ where $\sup\{k: c_k \neq 0\} < +\infty$, the

degree of P is just $\sup\{k: c_k \neq 0\}$.] Show that if $\sum_0^\infty a_k z^k = 0$ has no roots in common with $Q(z) = 0$, then the degree of P_k is exactly $n(c_k; Q)$.

Show that generally if $c_1, \dots, c_m$ are distinct points of $\mathbb{C}$, $d_1, \dots, d_\mu$ are distinct points of $\mathbb{C}$, $P_1, \dots, P_m$ and $Q_1, \dots, Q_\mu$ are polynomials of positive degree such that

$$P_1(0) = \dots = P_m(0) = Q_1(0) = \dots = Q_\mu(0) = 0$$

and

$$\sum_1^m P_k \left(\frac{1}{z - c_k} \right) = \sum_1^\mu Q_k \left(\frac{1}{z - d_k} \right), \quad z \in \mathbb{C}, \quad z \neq c_j, d_k,$$

then $\{c_1, \dots, c_m\} = \{d_1, \dots, d_\mu\}$ so that $\mu = m$. The map $j(k)$ of $\langle 1, m \rangle$ onto itself satisfying $d_k = c_{j(k)}$, $k = 1, \dots, m$, is such that $Q_k = P_{j(k)}$, $k = 1, \dots, m$.

1.14. Obtain the representation (1.4) for

$$f(z) = \frac{1}{(z-1)(z-2)^2(z-3)^3}.$$

Obtain the Laurent coefficients of $f|A(0; r_1, r_2)$ for the following (r_1, r_2) : $(1, 2)$, $(2, 3)$, $(3, +\infty)$.

1.15. Suppose that f is analytic on $\mathbb{C}$ and satisfies: (1) $f(z + 2\pi i) = f(z)$, $z \in \mathbb{C}$, (2) there exist positive numbers A and B such that $|f(z)| \leq Ae^{B|z|}$, $z \in \mathbb{C}$. [f satisfying (2) is said to be of *exponential type*.] Then $f = g \circ \exp$ for a g of the form $g(z) = \sum_{-\infty}^{\infty} a_k z^k$, where $n \in \mathbb{N}$ (Boas). Hint: Introduce

$$h = \{(e^z, f(z)): z \in \mathbb{C}\},$$

show that h is analytic on $\mathbb{C} - \{0\}$, show that its k th Laurent coefficient is given by

$$\frac{1}{2\pi} \int_0^{2\pi} f(x + yi) e^{-k(x+yi)} dy, \quad x \in \mathbb{R},$$

and estimate the size when $|k| > B$.

1.16. Let $(b_k): \mathbb{Z} \rightarrow \mathbb{C}$. Suppose that $c, d \in \mathbb{C}$ satisfy $0 < |c| < |d| < +\infty$ and that $\lim_{n \rightarrow \infty} \sum_{-n}^n b_k z^k$ exists and is in $\mathbb{C}$ when $z = c$ and when $z = d$. Show that the components of the Laurent pair $\sum_{\pm\infty} b_k z^k$ converge, respectively, when $|z| < |d|$ and when $|z| > |c|$.

1.17. Let $b_k = (|k| + 1)^{-2}$, $k \in \mathbb{Z}$. Show that the intersection of the convergence sets of the components of the Laurent pair $\sum_{\pm\infty} b_k z^k$ is $C(0; 1)$ and that the Laurent sum is continuous.

2. ISOLATED SINGULARITIES OF ANALYTIC FUNCTIONS

Let $a \in \mathbb{C}$ and let f be analytic in $\Delta(a; r) - \{a\}$ for some $r > 0$. Under this condition we say that f has an *isolated singularity* at a .† The Laurent sequence, $(b_k)_{k \in \mathbb{Z}}$, of $f|[\Delta(a; r) - \{a\}]$ is independent of $r > 0$ for which f is analytic in $\Delta(a; r) - \{a\}$. We term the common sequence in question the *a-Laurent*

† Of course, it is not excluded that f may be analytic at a .

sequence of f . We consider $\alpha = \inf\{k: b_k \neq 0\}$ and introduce the following definitions. We say that f has a *removable singularity at a* provided that $\alpha \geq 0$; that f has a (*polar singularity or*) *pole at a* provided that $-\infty < \alpha < 0$; that f has an *essential singularity at a* provided that $\alpha = -\infty$. When f has a pole at a , we term $|\alpha|$ the *order of the pole at a* . The first two possibilities are easily studied.

(1) *Removable singularity.* It is clear that f has a removable singularity at a only if there exists g analytic at a such that for some $r > 0$ we have $f(z) = g(z)$, for all z satisfying $0 < |z - a| < r$. The function f given by $f(0) = 1$, $f(z) = z$, $z \neq 0$, has a removable singularity at 0 as do z with domain $\mathbb{C}$ and z with domain $\mathbb{C} - \{0\}$.

(2) *Pole.* If f has a pole of order m at a , then when $r > 0$ is such that f is analytic in $\Delta(a; r) - \{a\}$, the series $\sum_0^\infty b_{-m+k}(z-a)^k$ converges when $|z-a| < r$. With g denoting its sum, we have

$$f(z) = \frac{g(z)}{(z-a)^m}, \quad 0 < |z-a| < r.$$

Hence $\lim_a f = \infty$ since $g(a) = b_{-m} \neq 0$.

(3) *Essential singularity.* We come to a famous theorem that gives information on the behavior of f near a when f has an essential singularity at a . It is the following theorem of Weierstrass.

Theorem 2.1: *If f has an essential singularity at a , then $\overline{f(V)} = \hat{\mathbb{C}}$ for every V contained in the domain of f which is of the form $U - \{a\}$, U a neighborhood of a .*

PROOF: Suppose that the assertion is false. Then there exist r positive, m positive, $A \in \mathbb{C}$, such that f is analytic in $\Delta(a; r) - \{a\}$ and $|f(z) - A| \geq m$ when $0 < |z-a| < r$. It is to be noted that if $\infty \notin \overline{f(V)}$, then $A \notin \overline{f(V)}$ for large $A \in \mathbb{C}$. By the Riemann theorem, Ex. 3.10, Ch. V (a second proof, Laurentian in character, is indicated in Ex. 2.1 below), we see that

$$[f(z) - A]^{-1} = (z-a)^n g(z), \quad (2.1)$$

$0 < |z-a| < r$, where $n \in \mathbb{N}$ and g is analytic on $\Delta(a; r)$ and does not take the value 0. Hence taking reciprocals of both sides of (2.1) we see that (b_k) , the a -Laurent sequence of f , has the property that $b_k = 0$, $k < -n$. The hypothesis is contradicted. The theorem follows.

Some remarks are in order. First, the conclusion of Theorem 2.1 is frequently given the following equivalent form: *For each $w \in \hat{\mathbb{C}}$, there exists $(z_k)_{k \in \mathbb{N}}$ such that $\lim_{k \rightarrow \infty} z_k = a$, each $z_k \neq a$ and is in the domain of f , $\lim_{k \rightarrow \infty} f(z_k) = w$.* It is routine to verify that the assertion just stated implies the conclusion of Theorem 2.1. In the other direction, we note that for n a sufficiently large positive integer $\Delta(a; n^{-1}) - \{a\}$ is contained in the domain of f and that for $w \in \mathbb{C}$ and such n there exists z_n satisfying $0 < |z_n - a| < n^{-1}$, $|f(z_n) - w| < n^{-1}$, and that for such n there exists z_n satisfying $0 < |z_n - a| < n^{-1}$, $|f(z_n)| > n$. Hence the conclusion of Theorem 2.1 implies the above assertion and equivalence follows.

Second, the propositions converse to those stated in (1) and (2) and to Theorem 2.1 hold because of the disjunction of the conclusions. (i) Thus, if g of (1) exists, then f has a removable singularity at a (as, of course, is immediate directly) since f cannot have a pole or essential singularity at a . The hypothesis of the Riemann theorem is a sufficient condition for a removable singularity, and we shall see in Ex. 2.1 below that a weaker hypothesis suffices. (ii) If $\lim_a f = \infty$, then f has a pole at a since, by (1) and Theorem 2.1, f cannot have a removable singularity nor an essential singularity at a . (iii) If f fulfills the conclusion of Theorem 2.1, then by (1) and (2), f cannot have a removable singularity or pole at a .

Third, the following condition, readily verified, is useful: if f has an isolated singularity at a and a is a cluster point of $\{f(z) = 0\}$, then either f has an essential singularity at a or $f(z) = 0$, $z \in \Delta(a; r) - \{a\}$, for some $r > 0$.

EXERCISES

2.1. Given f analytic in $\Delta(a; \rho) - \{a\}$, show with the aid of the Cauchy inequalities for the a -Laurent coefficients that if f does not have a removable singularity at a , then $\lim_0 r M(r)$ exists and is greater than 0 (possibly $+\infty$). Here

$$M(r) = \max \{|f(z)| : z \in C(a; r)\}.$$

The Riemann theorem is an immediate consequence. For a removable singularity the condition $\lim_{n \rightarrow \infty} r_n M(r_n) = 0$ suffices where (r_n) is a decreasing sequence of positive numbers satisfying $r_0 < \rho$, $\lim_{n \rightarrow \infty} r_n = 0$.

2.2. Let $f(z) = \exp(-z^{-2})$, $z \neq 0$. Show that f has an essential singularity at 0 and explain why this fact shows that the Taylor series about 0 of the function $F: \mathbb{R} \rightarrow \mathbb{R}$ given by $F(0) = 0$, $F(x) = \exp(-x^{-2})$, $x \neq 0$, which has derivatives of all orders on $\mathbb{R}$, cannot have sum $F|E$ on E where $0 < \inf E < +\infty$, $\inf E \notin E$.

2.3. Let $f(z) = \exp[(1-z)^{-1}]$, $z \neq 1$. Show that f has an essential singularity at 1 and determine its 1-Laurent sequence. Determine the Laurent coefficients of $f|A(0; 1)$ and of $f|A(0; 1, +\infty)$.

2.4. Let f be analytic in $\Delta(0; r) - \{1\}$, $r > 1$, and let f have a pole at 1. Let $\sum_0^\infty a_k z^k$ be the power series expansion of f in $\Delta(0; 1)$. Show that, for some $m \in \mathbb{N}$, we have $a_n \neq 0$ for $n \geq m$ and $\lim_{k \rightarrow \infty} a_{m+k+1}/a_{m+k} = 1$.

2.5. Let f and g be analytic on $\Delta(a; r)$, $r > 0$, and suppose that $f(a) \neq 0$, $g(a) = 0$, $g(z) \neq 0$ when $0 < |z - a| < r$. Let $h = f/g$. Then h has a pole of order $\nu = n(a; g)$ at a and the term b_k , $k \geq -\nu$, of the a -Laurent sequence of h is

$$\frac{1}{(\nu + k)!} \left(\frac{f}{g_1} \right)^{(\nu + k)}(a),$$

where g_1 is the function analytic on $\Delta(a; r)$ satisfying

$$g(z) = (z - a)^\nu g_1(z), \quad |z - a| < r.$$

Obtain the terms b_{-2} and b_{-1} of the 2-Laurent sequence of

$$(z - 1)^{-1}(z - 2)^{-2}(z - 3)^{-3}.$$

2.6. Let n denote an integer > 1 . Show with the aid of the Riemann criterion for a removable singularity that there does not exist a continuous n th root of z with domain $\Delta(0; r)$, $r > 0$ (or even $\Delta(0; r) - \{0\}$).

Isolated singularities at ∞ . It will be convenient to introduce the notation $\Delta(\infty; r)$ for the set $\{z: r^{-1} < |z| \leq +\infty\}$, where $r > 0$.† We are led, of course, to consider functions analytic in some $\Delta(\infty; r) - \{\infty\}$ and their behavior near ∞ . Parallel to what was said at the beginning of this section, a function f is said to have an *isolated singularity* at ∞ provided that it is analytic in some $\Delta(\infty; r) - \{\infty\}$. It is natural to consider the Laurent coefficients of $f|[\Delta(\infty; r) - \{\infty\}]$, r small and positive, for f having an isolated singularity at ∞ . Each Laurent coefficient b_k is independent of r for which f is analytic in $\Delta(\infty; r) - \{\infty\}$. We term the common sequence (b_k) the ∞ -Laurent sequence of f . We note that g defined by

$$g(z) = f(z^{-1}), \quad 0 < |z| < r, \quad (2.2)$$

r small positive, is analytic and (c_k) , the 0-Laurent sequence of g , satisfies $c_{-k} = b_k$, $k \in \mathbf{Z}$. On this basis we introduce $\beta = \sup\{k: b_k \neq 0\}$ and say that f has, respectively, a *removable singularity*, *pole*, *essential singularity at ∞* according as $\beta \leq 0$, $0 < \beta < +\infty$, $\beta = +\infty$. In the second case we term β the *order of the pole at ∞* . Given $f: U \rightarrow \hat{\mathbf{C}}$ where U is a neighborhood of ∞ , we say that f is *analytic at ∞* provided that f has a removable singularity and is continuous at ∞ . We extend the terminology “*analytic in (on) O* ” to open subsets O of $\hat{\mathbf{C}}$, in the first case meaning “analytic at each point” in the sense now available, and, in the second, requiring in addition that O be the domain of the function. It now follows that the statements made for a function having an isolated singularity at a point of $\mathbf{C}$ hold, *mutatis mutandis*, for a function having an isolated singularity at ∞ .

† We recall the definition $|\infty| = +\infty$.

EXERCISES

2.7. Verify the last assertion through the mediation of g of (2.2).

2.8. Suppose that f has an isolated singularity at ∞ . Let

$$M(r) = \max\{|f(z)|: |z| = r\},$$

r large and positive. Show that $\lim_{\infty} [\log M(r)] / \log r$ exists and that its possible values are in $\mathbb{N} \cup \{+\infty\}$. Further the limit is 0 if and only if f has a removable singularity at ∞ , it is a positive integer n if and only if f has a pole of order n at ∞ , it is $+\infty$ if and only if f has an essential singularity at ∞ .

2.9. Let f be analytic on $A(a; r, +\infty)$, let c_k denote the k th Laurent coefficient of f , and let (b_k) denote the ∞ -Laurent sequence of f . Show that

$$\sup\{k: b_k \neq 0\} = \sup\{k: c_k \neq 0\}.$$

Show that $c_{-1} = b_{-1}$ and determine c_{-k} in terms of $b_{-1}, \dots, b_{-k}$ for $k > 0$. What are the corresponding formulas for c_k , $k \geq 0$, when f has a pole at ∞ ?

2.10. Using the open mapping property of a nonconstant analytic function show that a function f analytic on $\mathbb{C}$ and univalent is a polynomial of degree 1, $f(z) = az + b$, $a \neq 0$. [It is immediate thereupon that $f(\mathbb{C}) = \mathbb{C}$.] Hint: Consider $f|[\Delta(\infty; 1) - \{\infty\}]$. The converse question is trivial.

2.11. Let $n \in \mathbb{N}$. Let

$$P(z) = z^{n+1} + \sum_0^n a_k z^k = \prod_1^m (z - \alpha_k)^{\nu_k},$$

the α_k being distinct and the ν_k being positive integers. Determine the ∞ -Laurent sequence (b_k) of P'/P in terms of the a_k and in terms of the α_k and ν_k . Show that $a_0, \dots, a_n$ may be represented in terms of $b_{-2}, \dots, b_{-(n+2)}$ and exhibit the determining equations.

2.12. Let f be analytic on $\mathbb{C}$. Let α denote the infimum of the set of nonnegative real numbers A satisfying the condition that there exists $r \in \mathbb{R}^+$ such that $|f(z)| \leq e^{A|z|}$, $|z| > r$. When $\alpha < +\infty$, we say that f is of *exponential type* α . (1) Show that if g is analytic on $\{|z| > r\}$ and γ is a rectifiable closed curve in $A(0; r, +\infty)$, then

$$f(z) = \int_{\gamma} e^{zw} g(w) dw, \quad z \in \mathbb{C},$$

is of exponential type $\alpha \leq r$. (2) With the aid of the Cauchy inequalities show that if f is of exponential type α and $\alpha < \beta < +\infty$, then $a_k = f^{(k)}(0)/k!$ satisfies

$$|a_k| \leq \left(\frac{\beta e}{k}\right)^k$$

for large k and conclude that the radius of convergence ρ of $\sum_0^{\infty} k! a_k z^k$ satisfies $\rho \geq (\alpha e)^{-1}$ and that the sum F of this series has the property that

$$f(z) = \frac{1}{2\pi i} \oint_{C(0; r)} e^{zw} \frac{1}{w} F\left(\frac{1}{w}\right) dw,$$

where $1/\rho < r < +\infty$. What is F , when f is $\exp$? $\cos$?

2.13. Let $\lambda \in \mathbb{C}$. Show that if $2\lambda \notin \mathbb{Z}$, then the differential equation

$$z w''(z) + (2\lambda + 1) w'(z) + z w(z) = 0 \quad (2.3)$$

(which is closely allied to the Bessel equation of order λ) admits solutions other than the constant 0 analytic on $\mathbb{C}$ and that they are mutually proportional. The method of undetermined coefficients is to be used. Show that the solutions in question are of exponential type 1. Determine the solutions of (2.3) analytic on $\mathbb{C} - \{0\}$, no restriction being placed on λ . Show that, if $2\lambda \notin \mathbb{Z}$, they have removable singularities at 0 and are proportional. What happens when $2\lambda \in \mathbb{Z}$?

Principal part. Suppose that f has an isolated singularity at $a \in \hat{\mathbb{C}}$. Let (b_k) denote the a -Laurent sequence of f . When $a \in \mathbb{C}$, we let g denote the sum on $\mathbb{C}$ of $\sum_1^\infty b_{-k} z^k$ and term $g[(z-a)^{-1}]$ with domain $\hat{\mathbb{C}} - \{a\}$ the *principal part of f at a* . In this case the principal part of f at a is analytic at ∞ . When $a = \infty$, we define the *principal part of f at ∞* to be the sum on $\mathbb{C}$ of $\sum_1^\infty b_k z^k$.

EXERCISE

2.14. Determine the principal part of $\exp$ at ∞ , at 0. Determine the principal part of e^{-1/z^2} at 0; of $\exp(z + z^{-1})$ at 0; of $(z-1)^{-3}e^z$ at 1; of $\sin[(1-z)^{-1}]$ at 1, at ∞ .

Partial fraction decomposition. We may now rapidly complete the study of the partial fraction decomposition of a rational function. Let P and Q be polynomials of respective degrees m and n , each nonnegative. Let $P(z) = \sum_0^m a_k z^k$ and $Q(z) = \sum_0^n b_k z^k$. We have

$$\frac{P(z)}{Q(z)} = \frac{a_m}{b_n} z^{m-n} \varphi(z),$$

z large and finite, where $\lim_{\infty} \varphi = 1$. Hence $P(z)/Q(z)$ has a limit at ∞ and, consequently, a removable singularity or pole at ∞ . Let $\alpha_1, \dots, \alpha_v$ denote the distinct roots of $Q(z) = 0$, let g_k denote the principal part of $P(z)/Q(z)$ at α_k and let g_∞ denote the principal part of $P(z)/Q(z)$ at ∞ . The function

$$\frac{P(z)}{Q(z)} - g_\infty(z) - \sum_1^v g_k(z),$$

with domain $\mathbb{C} - \{\alpha_1, \dots, \alpha_v\}$ has removable singularities at $\alpha_1, \dots, \alpha_v$ and ∞ . By the theorem of Liouville it is constant. As we see easily, the value of the constant is the term b_0 with index 0 of the ∞ -Laurent sequence of $P(z)/Q(z)$. Hence

$$\frac{P(z)}{Q(z)} = b_0 + g_\infty(z) + \sum_1^v g_k(z).$$

EXERCISES

2.15. Show that if the equations $P(z) = 0$ and $Q(z) = 0$ have no roots in common, then g_k has a pole of order $n(\alpha_k; Q)$ at α_k , $k = 1, \dots, \nu$, and g_∞ is a polynomial of degree $m - n$ when $m > n$ and otherwise is 0. What is the situation when there are common roots?

2.16. Show directly from Laurent series considerations that if f is analytic on $\hat{\mathbb{C}}$, then f is constant.

3. RESIDUE

Let f have an isolated singularity at a . By the *residue of f at a* , denoted $\text{res}(a; f)$, is meant b_{-1} , the term of the a -Laurent sequence of f with index -1 , when $a \in \mathbb{C}$, and is meant $-b_{-1}$ when $a = \infty$. The definition of $\text{res}(\infty; f)$ may appear a little surprising, but the factor -1 will be seen to be very natural, and indeed it renders certain statements more symmetric. The notion of residue is useful for it focuses attention on the one term of the Laurent expansion which may make its presence felt under integration along a rectifiable closed curve not passing through the point a . We have already seen the role played by the term b_{-1} of a Laurent sequence in the study of the existence of a primitive.

EXERCISE

3.1. Calculate the residue of $(\cot \pi z)^{2n+1}$ at 0, $n \in \mathbb{N}$. Hint: Consider the integral of $(\cot \pi z)^{2n+1}$ along the positively sensed boundary path of the rectangle $[-\frac{1}{2}, \frac{1}{2}; -T, T]$, T positive, and study the limiting behavior of the integral as a function of T at $+\infty$ [cf. 117, p. 108]. The point of this example is that, although we use the knowledge of a residue to evaluate certain definite integrals, there are occasions such as the present one where direct attack on the Laurent expansion would be quite forbidding while the skillful choice of an integral representation of the residue makes the calculation very simple. For another example of this kind, cf. the very ingenious treatment of Gaussian sums due to Mordell, given in §1, Ch. XIV, of this book.

We consider a residue theorem for analytic functions on $\mathbb{C}$ less a finite number of points.

Theorem 3.1: Let $a_1, \dots, a_n \in \mathbb{C}$ and let f be analytic on $\mathbb{C} - \{a_1, \dots, a_n\}$. Then

$$\text{res}(\infty; f) + \sum_{k=1}^n \text{res}(a_k; f) = 0.$$

PROOF: We note that

$$f(z) = c + g_{\infty}(z) + \sum_1^n g_k(z),$$

where $c \in \mathbb{C}$, g_{∞} is the principal part of f at ∞ , and g_k is the principal part of f at a_k . Now

$$\text{res}(a_k; f) = \text{res}(a_k; g_k), \quad k = 1, \dots, n,$$

and

$$\text{res}(\infty; f) = \sum_1^n \text{res}(\infty; g_k).$$

The theorem follows on noting that $\text{res}(a_k; g_k) + \text{res}(\infty; g_k) = 0$ as we see either by representing the residues as integrals of g_k along $\Gamma(r) = re^{2\pi it}$, r large and positive, or else by using the easily verified

$$\begin{aligned} \text{res}(a_k; g_k) &= \lim_{\infty} (z - a_k) g_k(z) \\ &= \lim_{\infty} \left(1 - \frac{a_k}{z}\right) z g_k(z) = -\text{res}(\infty; g_k). \end{aligned}$$

EXERCISES

3.2. Let $n \in \mathbb{N}$, $a, b \in \mathbb{C}$. Let

$$f(z) = z(z - a)^{-(n+2)}(z - b)^{-1}.$$

Calculate $\text{res}(a; f)$, $\text{res}(b; f)$.

3.3. Show that f of Theorem 3.1 admits a primitive if and only if $\text{res}(a_k; f) = 0$, $k = 1, \dots, n$.

3.4. Let P and Q be polynomials of degrees m and n , respectively, $0 \leq m \leq n$. It is supposed that $P(z) = 0$ and $Q(z) = 0$ have no roots in common. Suppose further that $f(z) = P(z)/Q(z)$ and that $f(z) - z = 0$ and $f'(z) = 1$ have no roots in common. Show that $\sum_{a \in E} [1 - f'(a)]^{-1} = 1$, where $E = \{f(z) = z\}$.

3.5. At this stage it is desirable to have practice in specifying the nature of isolated singularities and determining the associated residues. This is particularly important for the calculation of integrals via Cauchy's residue theorem to be treated in Ch. VII. State the nature of the isolated singularity and determine the value of the residue at each point $a \in \hat{\mathbb{C}}$ such that the specifying rule defines an analytic function in $\Delta(a; r) - \{a\}$ for some $r > 0$. Exercise 2.5, this chapter, will be found useful.

(a) $z^{-(n+1)}e^{\alpha z}$, $n \in \mathbb{N}$, $\alpha \in \mathbb{C}$. (b) $(1 + z^{n+1})^{-1}$, $n \in \mathbb{N}$. (c) $z^m(1 + z^{2n})^{-1}$, $m, n \in \mathbb{N}$. (d) $\sin \alpha z/[z(z^2 + 1)]$, $\alpha \in \mathbb{C}$. (e) $\tan \pi z$. (f) $\csc \pi z = (\sin \pi z)^{-1}$. (g) $(\cosh z)^{-1}$ where $\cosh z = (e^z + e^{-z})/2$. (h) $(1 + z^2)^{-(n+1)}$, $n \in \mathbb{N}$. (i) $e^{\alpha z}/(1 + e^z)$, $\alpha \in \mathbb{C}$. (j) $f(z)(1 + z)^{-1}$ where f is the analytic α th power of the identity map on $\mathbb{C} - \mathbb{R}^+$ specified by the

condition that $f(z) = \exp(\alpha \operatorname{Log} z)$, $\operatorname{Im} z > 0$. Here $\alpha \in \mathbb{C}$. (k) $(\operatorname{Log} z)(1 - z)^{-3}$. (l) Let P be a polynomial of positive degree. Determine the requested information for $P'(z)/P(z)$. (m) Given f analytic on $\Delta(0; r)$ and P as in (l), determine the requested information for $f(z)/P(z)$ under the assumption that $P(z) = 0$ has only *simple* roots, that is, when $P(z) = 0$, $n(z; P) = 1$.

An invariance property of the residue. Actually, the notion of a residue is best formulated in terms of differentials on Riemann surfaces. This approach goes beyond our immediate program (cf. Ch. XVIII). However, we can bring to light at this stage the action of analytic changes of variable on the residue which bears directly on the Riemann surface aspect of the question. Let $a, A \in \mathbb{C}$, let f be analytic on $\Delta(A; r) - \{A\}$, and let φ denote the restriction to $\Delta(a; s) - \{a\}$ of $\psi: \Delta(a; s) \rightarrow \Delta(A; r)$, $\psi(a) = A$, ψ analytic on $\Delta(a; s)$ and univalent. We have

Theorem 3.2: $\operatorname{res}(A; f) = \operatorname{res}[a; (f \circ \varphi)\varphi']$.

PROOFS: (1) Using integrals. Clearly

$$\operatorname{res}[a; (f \circ \varphi)\varphi'] = \frac{1}{2\pi i} \int_{\varphi \circ \Gamma_\rho} f dz, \quad 0 < \rho < s,$$

where $\Gamma_\rho(t) = a + \rho e^{2\pi i t}$. We note that

$$\varphi(z) - A = (z - a) \varphi_1(z),$$

where φ_1 is analytic on $\Delta(a; s)$ and does not take the value 0. Hence

$$O(A; \varphi \circ \Gamma_\rho) = O(a; \Gamma_\rho) = 1$$

and we conclude by Ex. 5.7, Ch. V, and the homotopic version of the Cauchy integral theorem that

$$\operatorname{res}[a; (f \circ \varphi)\varphi'] = \frac{1}{2\pi i} \oint_{C(A; \rho)} f dz, \quad 0 < \rho < r,$$

and hence the assertion of the theorem.

(2) Using a Laurentian approach. Let (b_k) be the A -Laurent sequence of f . Let g denote a primitive of

$$f(z) - b_{-1}(z - A)^{-1}.$$

We have

$$f[\varphi(z)] \varphi'(z) = (g \circ \varphi)'(z) + b_{-1} \frac{\varphi'(z)}{\varphi(z) - A},$$

$0 < |z - a| < s$. Since $(g \circ \varphi)'$ is a derivative, $\text{res}[a; (g \circ \varphi)'] = 0$. Clearly $\text{res}[a; \varphi' / (\varphi - A)] = 1$. The assertion follows.

The case $a, A \in \hat{\mathbb{C}}$, will be treated in Ex. 4.16 of the next section.

4. MEROMORPHIC FUNCTIONS

From certain points of view it is natural and appropriate to assign the value ∞ at a pole. Continuity (in the sense of the topology of $\hat{\mathbb{C}}$) is thereby achieved. In the present section we shall see that a harmonious concept—that of a meromorphic function—results. We shall proceed from a formal definition to a study of the most immediate properties of meromorphic functions. The program carried out for analytic functions suggests not only a list of questions to be considered at once—such as those pertaining to algebraic structure, behavior under composition, inverse, and limit problems—but also, as we should expect, most of the basic tools. In the study of limit problems a note of great simplicity is achieved by the introduction of the chordal metric on $\hat{\mathbb{C}}$ which has a patent geometric significance. The topology it induces is just the standard topology of $\hat{\mathbb{C}}$.

Let $f: V \rightarrow \hat{\mathbb{C}}$ where V is a neighborhood of $a \in \hat{\mathbb{C}}$. We say that f is *meromorphic at a* provided that f is continuous at a and has an isolated singularity at a . Let O be an open subset of $\hat{\mathbb{C}}$. We say that f is *meromorphic in O* provided that f is meromorphic at each point of O , and that f is *meromorphic on O* provided that O is the domain of f and f is meromorphic in O . We denote the set of functions meromorphic on O by $\mathbf{M}(O)$. It is to be observed that our use earlier of maps of subsets of $\hat{\mathbb{C}}$ into $\hat{\mathbb{C}}$ when we studied analytic functions looked forward to the introduction of the notion of a meromorphic function.

EXERCISES

4.1. Given $f: A \rightarrow \hat{\mathbb{C}}$ where A is a region in $\hat{\mathbb{C}}$. Then f is meromorphic on A if and only if f is not the constant with value ∞ and for each $a \in A$ there exists $\Delta(a; r) \subset A$ such that either f or $1/f$ is analytic in $\Delta(a; r)$. Why is the problem stated with “region” rather than “open set”?

4.2. Given $f: O \rightarrow \hat{\mathbb{C}}$, where O is an open subset of $\hat{\mathbb{C}}$. Suppose that f has an isolated singularity at each point of O but does not have an essential singularity at any point of O . Show that there exists a unique g meromorphic on O such that $\{f(z) \neq g(z)\}$ clusters at no point of O .

4.3. Suppose that f is meromorphic on $\Delta(a; r) - \{a\}$. Show that either f is the restriction to $\Delta(a; r) - \{a\}$ of a function meromorphic on $\Delta(a; r)$ or else when $0 < s < r$, $f[\Delta(a; s) - \{a\}]$ is dense in $\hat{\mathbb{C}}$ (cf. the Weierstrass theorem, Theorem 2.1, this chapter). Show that a sufficient condition for the second alternative is that f be not constant and $\{f(z) = A\}$ cluster at a for some $A \in \hat{\mathbb{C}}$.

4.4. Let f be meromorphic on $\mathbb{C}$ and satisfy $f(z + 2\pi i) = f(z)$, $z \in \mathbb{C}$. Show with the aid of Ex. 4.3 that $f = g \circ \exp$ for some g meromorphic on $\hat{\mathbb{C}}$ if and only if $\lim_{\operatorname{Re} z \rightarrow +\infty} f$ and $\lim_{\operatorname{Re} z \rightarrow -\infty} f$ exist. Here the obvious gloss is to be made: $\lim_{\operatorname{Re} z \rightarrow +\infty} f = a \in \hat{\mathbb{C}}$ provided that for each neighborhood U of a there exists $b \in \mathbb{R}$ such that $f(z) \in U$ when $\operatorname{Re} z \geq b$, the other limit being similarly defined with “ $-\infty$ ” replacing “ $+\infty$ ” and “ $\leq$ ” replacing “ $\geq$.”

Divisor. Given f meromorphic at a we define $\partial_f(a)$, the divisor of f at a , as $\inf\{k: b_k \neq 0\}$, where (b_k) is the a -Laurent sequence of f when $a \neq \infty$, and as $\inf\{-k: b_k \neq 0\}$ when $a = \infty$. Given f meromorphic on O , an open subset of $\hat{\mathbb{C}}$, by the divisor of f , ∂_f , is meant the function with domain O assigning the value $\partial_f(a)$ to $a \in O$. It is a useful and convenient notion.

Suppose that f is meromorphic on a region $A \subset \hat{\mathbb{C}}$. Let E denote the set of points for which $\{f(z) = 0\}$ is a neighborhood. Then, as in the case of analytic functions (cf. Ex. 6.7, Ch. IV), either $E = A$ so that f is the constant 0, or else the set $\{f(z) = 0\}$ clusters at no point of A . We now see that if $c \in \mathbb{C}$, then either f is the constant function taking the value c or else $\{f(z) = c\}$ clusters at no point of A . It suffices to consider $g(z) = f(z) - c$. We also note that when f is not the constant zero, $1/f$ is meromorphic on A .

Algebraic properties. Given f, g meromorphic on O , an open subset of $\hat{\mathbb{C}}$, we let

$$E = \{f(z) \neq \infty, g(z) \neq \infty\}.$$

We note that $(f|E) + (g|E)$ and $(f|E)(g|E)$ are analytic on E , and, as follows from examining their a -Laurent sequences, each is the restriction of a unique function meromorphic on O . We denote by $f + g$ the unique meromorphic function on O whose restriction to E is $(f|E) + (g|E)$ and term it the *sum of f and g* . Similarly, we denote by fg the unique meromorphic function on O whose restriction to E is $(f|E)(g|E)$ and term it the *product of f and g* . Given $\alpha \in \mathbb{C}$ by αf we understand the product of the constant on O taking the value α and f .

EXERCISES

4.5. Show that when A is a region in $\hat{\mathbb{C}}$, $\mathbf{M}(A)$ together with addition and multiplication as just defined constitutes a field. What are its zero and identity? Show that if $f \in \mathbf{M}(A)$ is not the zero element, then $1/f$ is the multiplicative inverse of f . What complications arise when we consider $\mathbf{M}(O)$, where O is not connected?

4.6. Given $f \in \mathbf{M}(A)$, f not the zero constant. Then $\{\partial_f(z) \neq 0\}$ clusters at no point of A . The fact that A is a region is essential.

4.7. Given $f, g \in \mathbf{M}(O)$. Show that

$$\partial_{f+g} \geq \min\{\partial_f, \partial_g\}, \quad \partial_{fg} = \partial_f + \partial_g.$$

[Given $\varphi, \psi: X \rightarrow \hat{\mathbb{R}}$, by $\min\{\varphi, \psi\}$ we understand the map with domain X assigning to $x \in X$ the value $\min\{\varphi(x), \psi(x)\}$.]

Derivative. Let f be meromorphic on an open subset O of $\hat{\mathbb{C}}$. Let $E = \{\infty\} \cup \{f(z) = \infty\}$. Then there is a unique meromorphic function on O whose restriction to $O - E$ is $[f|_{(O - E)}]'$. We term it the *derivative of f* and denote it by f' . The notion is, of course, consistent with that of the earlier analytic theory. The derivative f' is characterized by the property that for each $a \in O$ the a -Laurent sequence of f' is just $((k+1)b_{k+1})$ where (b_k) is the a -Laurent sequence of f . It is clear that the usual rules for the derivative of a sum and product hold.

EXERCISE

4.8. Given f meromorphic on a region A , f not the constant zero. Show that $\text{res}(z; f'/f) = \partial_r(z)$, $z \in A$.

Local analysis of a meromorphic function. Suppose that f is meromorphic at a . Let $f(a) = A$. Let $\varphi_c(z) = z + c$ if $c \neq \infty$, but $1/z$ if $c = \infty$. Let $\psi_c(z) = z - c$ if $c \neq \infty$, but $1/z$ if $c = \infty$. The function

$$g(z) = \psi_A\{f[\varphi_a(z)]\}, \quad (4.1)$$

with domain $\Delta(0; r)$, r small and positive, is continuous at 0, takes the value 0 at 0, and is analytic in $\Delta(0; r) - \{0\}$. Hence g is analytic on $\Delta(0; r)$. When $a \neq \infty$, we let μ denote the absolute value of

$$\inf\{k: k \neq 0, b_k \neq 0\},$$

and when $a = \infty$, we let μ denote the absolute value of

$$\sup\{k: k \neq 0, b_k \neq 0\},$$

(b_k) being in either case the a -Laurent sequence of f . It is a matter of straightforward verification to check that $\mu = n(0; g)$, the multiplicity of g at 0. We define the *multiplicity of f at a* , denoted $n(a; f)$, to be μ . In case $a, b \in \mathbb{C}$, the definition is consistent with the one given earlier for analytic functions.

EXERCISES

4.9. Let h be meromorphic at b , and let f be meromorphic at a , satisfy $f(a) = b$, and be not constant in any neighborhood of b . Let $F(z) = h[f(z)]$, z near a . Then F is meromorphic at a and

$$n(a; F) = n(a; f) n(b; h).$$

Hint: Introduce the g associated with f and h .

4.10. Verify that the chain rule holds for the composition of meromorphic functions with an obvious proviso.

The *valence* of f meromorphic on a region is defined exactly as the valence of a function analytic on a region with multiplicity being taken in the sense of meromorphic functions. We note that from (4.1) we obtain

$$f(z) = \varphi_A\{g[\psi_a(z)]\},$$

z near a . Combining this relation with the corollary of Theorem 9.2 of Ch. V we conclude: Let $F = f| \Delta(a; r)$, where r is small and positive. Then $v_F(w) \leq n(a; f)$, $w \in \hat{\mathbb{C}}$. Further, $v_F(w) = n(a; f)$ for w sufficiently near A when $n(a; f) < +\infty$.

EXERCISES

4.11. Furnish the details of the proof of the last assertion.

4.12. Show that if f is meromorphic on a region and is not constant, then f has the open mapping property.

4.13. Show that if f is meromorphic on a region A and is univalent, then the inverse of f is meromorphic on $f(A)$.

4.14. Let f be meromorphic on a region $A \subset \hat{\mathbb{C}}$. Let $g: B \rightarrow A$ where B is a region in $\hat{\mathbb{C}}$, and g is meromorphic on B and is not constant. Let $h = f \circ g$. Show that

$$\partial_h(z) = \partial_f[g(z)] n(z; g), \quad z \in B.$$

4.15. Local equivalence. Given that f is meromorphic at a and g is meromorphic at b , we say that (f, a) is *locally equivalent* to (g, b) provided that there exist φ meromorphic at a , ψ meromorphic at $g(b)$ satisfying $\varphi(a) = b$, $\psi[g(b)] = f(a)$,

$$n(a; \varphi) = n[g(b); \psi] = 1,$$

such that $f(z) = \psi\{g[\varphi(z)]\}$, z near a . Show that an equivalence relation is defined thereby and that (f, a) is locally equivalent to (g, b) if and only if either (1) f is constant in a neighborhood of a and g is constant in a neighborhood of b or (2) neither f is constant in a neighborhood of a nor g in a neighborhood of b , and in addition $n(a; f) = n(b; g)$. In particular, if f is not constant in a neighborhood of a , (f, a) is locally equivalent to $(z^n, 0)$ for exactly one positive integer m , namely $n(a; f)$. [The notion of an equivalence relation is recalled in §3, Ch. X.]

4.16. Behavior of residue under composition. Let g have an isolated singularity at b and let f be meromorphic at a and satisfy: $f(a) = b$, f is not constant at a . Let

$$h(z) = g[f(z)] f'(z), \quad z \in \Delta(a; r) - \{a\},$$

r small positive. Show that

$$\text{res}(a; h) = \text{res}(b; g) n(a; f).$$

4.17. Given that f is meromorphic on $\Delta(a; r) - \{a\}$ and is not constant. Show that if v_r is bounded on some nonempty open set O of $\hat{\mathbb{C}}$, then f is the restriction of a function meromorphic on $\Delta(a; r)$. Apply to study f meromorphic on $\mathbb{C}$, not constant,

with valence bounded on some nonempty open set. Hint: Consider such an O and $b \in O$ at which $v_f|O$ is maximized, and apply the valence property of f restricted to small disks centered at the points $c, f(c) = b$.

4.18. Given f analytic on $\mathbb{C}$, not a polynomial (such a function is called a *transcendental entire* function), and g analytic on $\Delta(a; r) - \{a\}$. Suppose that $\lim_a f \circ g$ exists (in $\hat{\mathbb{C}}$). Then g has a removable singularity at a .

Argument principle. Let A be a holomorphically simply connected region in $\mathbb{C}$, let γ be a closed curve in A , let f be meromorphic on A and suppose that $f(z) \neq 0, \infty, z \in \gamma$. Suppose further that $v_f(0) + v_f(\infty) < +\infty$. We seek to determine the influence of the zeros and poles of f on $O(0; f \circ \gamma)$, a zero of a function meromorphic on a region being a point at which the function takes the value zero. More accurately, we seek to relate ∂_f and $O(0; f \circ \gamma)$. We note that

$$f = \left[\prod_{\partial_f(a) \neq 0} E_a^{\partial_f(a)} \right] g$$

where $E_a(z) = z - a$ and g is analytic on A and does not take the value 0. What was said earlier (Ex. 3.1, Ch. V) about sums over finite sets applies, *mutatis mutandis*, to products. We conclude that

$$O(0; f \circ \gamma) = \left[\sum_{\partial_f(a) \neq 0} O(0; E_a \circ \gamma) \partial_f(a) \right] + O(0; g \circ \gamma).$$

Now $O(0; g \circ \gamma) = 0$ since g has an analytic logarithm and hence the continuous logarithms of $g \circ \gamma$ are periodic with period 1. Hence we have the following equality, which the *argument principle* states under the above hypotheses,

$$O(0; f \circ \gamma) = \sum_{\partial_f(a) \neq 0} O(a; \gamma) \partial_f(a).$$

It is to be noted that in case γ is rectifiable we have available the integral representation

$$O(0; f \circ \gamma) = \frac{1}{2\pi i} \int_{\gamma} \frac{f'}{f} dz.$$

Valence and cluster values. Let f be meromorphic on a region $A \subset \hat{\mathbb{C}}$. By a *cluster value* of f is meant a member of the set

$$\bigcap \overline{f(A - C)},$$

the intersection being taken with respect to C compact $\subset A$.† We see that in

† Here, of course, we could consider C closed in the sense of $\hat{\mathbb{C}}$ which are contained in A . The choice of the term “compact” is motivated by regard for the general theory.

case $A = \mathbb{C}$, for a given f meromorphic on $\mathbb{C}$, the set of cluster values of f is either $\hat{\mathbb{C}}$ or else reduces to a singleton (cf. Ex. 4.3, this section). The relation between cluster values and valence is brought to light by the following theorem.

Theorem 4.1: *Given f meromorphic on a region $A \subset \hat{\mathbb{C}}$. A point $b \in \hat{\mathbb{C}}$ is not a cluster value of f if $v_f(b) < +\infty$ and $\{v_f(z) = v_f(b)\}$ is a neighborhood of b . If $A \neq \hat{\mathbb{C}}$ or f is not constant, then the converse statement holds; however it is false when f is a constant on $\hat{\mathbb{C}}$.*

PROOF: Let us consider the first assertion. It is obvious when $v_f(b) = 0$. If $v_f(b) > 0$, let $a_1, \dots, a_m$ denote the points of A where f takes the value b , $a_k \neq a_j$ for $k \neq j$. Let $r > 0$ be chosen so that $\overline{\Delta(a_k; r)} \subset A$, $k = 1, \dots, m$, and the $\overline{\Delta(a_k; r)}$ are mutually disjoint. The valence of $f|_{\Delta(a_k; r)}$ takes values at least as large as $n(a_k; f)$ in some neighborhood of b . Hence there exists a neighborhood of b contained in the complement of

$$f\left[A - \bigcup_{1 \leq k \leq m} \overline{\Delta(a_k; r)}\right]$$

and consequently b is not a cluster value of f .

When f is constant and $A \neq \hat{\mathbb{C}}$, there is just one cluster value of f , the constant value c of f . The valence of f takes the value 0 on $\hat{\mathbb{C}} - \{c\}$ and the second assertion follows in this case. Suppose that f is not constant and that b is not a cluster value of f . The set $\{f(z) = b\}$ is finite, otherwise $b \in f(A - C)$ for each compact $C \subset A$. Let $a_1, \dots, a_m$ denote the members of $\{f(z) = b\}$, $a_k \neq a_j$ for $k \neq j$. Let positive r be so chosen that the $\overline{\Delta(a_k; r)}$ are mutually disjoint and lie in A and that the valence of $f|_{\Delta(a_k; r)}$ does not take values greater than $n(a_k; f)$. Now

$$b \notin f\left[A - \bigcup_{1 \leq k \leq m} \overline{\Delta(a_k; r)}\right].$$

Otherwise, there would exist $(z_n)_{n \in \mathbb{N}}$, $\lim_{n \rightarrow \infty} z_n = a$, $z_n \in A - \bigcup_{1 \leq k \leq m} \overline{\Delta(a_k; r)}$ for all n , $\lim_{n \rightarrow \infty} f(z_n) = b$. If $a \in A$, then $a \neq a_k$ and the assumption concerning $a_1, \dots, a_m$ would be violated, while if $a \notin A$, then b would be a cluster value since then b would belong to each $f(A - C)$. Consequently, there exists s positive such that

$$f^{-1}[\Delta(b; s)] \subset \bigcup_{1 \leq k \leq m} \Delta(a_k; r)$$

and we conclude that

$$v_f(w) \leq \sum_1^m n(a_k; f) = v_f(b)$$

for $w \in \Delta(b; s)$ and that $v_f(w) = v_f(b)$ for w sufficiently near b . The proof follows.

When f is constant and $A = \hat{C}$, there are no cluster values but the valence is $+\infty$ at the constant value of f and hence the converse statement does not hold in this case.

Theorem 4.1 has the following consequence: If $f \in M(\hat{C})$ is not constant, then v_f is locally constant, for the set of cluster values is empty. By the connectedness of $\hat{C}$, v_f is constant. Of course the constant value is a positive integer.

EXERCISES

4.19. Given f meromorphic on a region $A \subset \hat{C}$. $w \in \hat{C}$ is a cluster value of f if and only if there exists $(z_n): \mathbb{N} \rightarrow A$ satisfying: (z_n) possesses a limit in $\hat{C} - A$, $\lim_{n \rightarrow \infty} f(z_n) = w$.

4.20. Show that one may replace a region by an unrestricted open set in the above study relating cluster values and valence.

4.21. Given $f \in M(\hat{C})$, not constant. Show that there exist polynomials P, Q of respective degrees m, n , $m \geq 0, n \geq 0$, having no zeros in common such that $f(z) = P(z)/Q(z)$, $z \in C$. Show directly without appeal to the above developments of the text that $v_f(w) = \max\{m, n\}$, $w \in \hat{C}$. To what extent are P and Q determined?

4.22. Let $n \in \mathbb{N}$, let $a_k, b_k \in \mathbb{R}$, $k = 0, \dots, n+1$, $a_{n+1} + ib_{n+1} \neq 0$. Let

$$f(z) = \sum_{k=0}^{n+1} (a_k \cos kz + b_k \sin kz).$$

Show that $f(z) = g(e^{iz})$ where $g(z) = \sum_{k=-(n+1)}^{n+1} A_k z^k$, $A_k = A_{-k}$, $A_{n+1} \neq 0$, and conclude that the value 0 is attained by f on $\{-\pi \leq \operatorname{Re} z < \pi\}$ exactly $2(n+1)$ times, counted according to multiplicity. Is the theorem true when we allow $n = -1$?

Theorem of Rouché. There are certain situations where Theorem 4.1 yields useful information. Thus when $A \neq \hat{C}$, then v_f is finite and constant on each component of the complement with respect to $\hat{C}$ of the set of cluster values K . In particular, when $K \subset \Delta(1; 1)$, then 0 and ∞ are in the same component of $\hat{C} - K$, and $v_f(0) = v_f(\infty) < +\infty$. This simple remark, extended to the case where A is replaced by O open, permits an easy proof of the following theorem of Rouché.

Theorem 4.2: Given O a proper open subset of $\hat{C}$, let $f, g: \bar{O} \rightarrow \hat{C}$ be continuous on $\bar{O}$, take values in C on $\operatorname{fr} O$, and satisfy $|f(z)| < |g(z)|$, $z \in \operatorname{fr} O$. Further, suppose that f and g are meromorphic in O . Let h be the unique continuous map of $\bar{O}$ into $\hat{C}$ satisfying $h|E = (f|E) + (g|E)$ where

$$E = \{f(z) \neq \infty, g(z) \neq \infty\}.$$

Then

$$\nu_{g|O}(0) - \nu_{g|O}(\infty) = \nu_{h|O}(0) - \nu_{h|O}(\infty).$$

(The values of the valences entering into the equality are finite.)

PROOF: The finiteness of the valences in question is a consequence of the fact that $g(z), h(z) \in \mathbb{C} - \{0\}$ when $z \in \text{fr } O$. If either $\{g(z) = 0\}$ or $\{g(z) = \infty\}$ were infinite, then g would take the value 0 or ∞ somewhere on $\text{fr } O$. A similar remark holds for h . We note that the set of cluster values of $(h|O)/(g|O)$ is exactly

$$\left\{ 1 + \frac{f(z)}{g(z)} : z \in \text{fr } O \right\}. \quad (4.2)$$

Indeed, in general, if $\varphi: \bar{O} \rightarrow \hat{\mathbb{C}}$ is continuous on $\bar{O}$ and meromorphic in O , then the set of cluster values of $\varphi|O$ is precisely $\varphi(\text{fr } O)$ (cf. Ex. 4.19, this section). Since the set (4.2) is contained in $\Delta(1; 1)$, we see that if ∂ denotes the divisor of $(h|O)/(g|O)$, then

$$0 = \sum_{\partial(z) \neq 0} \partial(z) = \sum \partial_{h|O}(z) - \sum \partial_{g|O}(z),$$

the latter two sums being taken over the zeros and poles of g and of h . The assertion of the theorem follows when we note that $\sum \partial_{h|O}(z)$ is just $\nu_{h|O}(0) - \nu_{h|O}(\infty)$ and that a corresponding statement holds for g .

EXERCISES

- 4.23.** Let $n \in \mathbb{N}$. Then $z^{n+3} + e^z$ has $n+3$ zeros counted by multiplicity in $\Delta(0; e)$.
4.24. Let O be as in Theorem 4.2. Let F be a continuous map of $\bar{O} \times [0, 1]$ into $\hat{\mathbb{C}}$ such that F takes values in $\mathbb{C} - \{0\}$ at points of $(\text{fr } O) \times [0, 1]$. For each $t \in [0, 1]$ let f_t , the function with domain O given by $f_t(z) = F(z, t)$, be meromorphic on O . Then $\nu_{f_t}(0) - \nu_{f_t}(\infty)$ is constant. Does one need the full force of the hypothesis on F ? Does such a theorem hold when $[0, 1]$ is replaced by a connected compact metric space?

We return to the study of functions meromorphic on $\hat{\mathbb{C}}$. They are exactly the continuous functions on $\hat{\mathbb{C}}$ whose restrictions to $\mathbb{C}$ are the quotients of polynomials P/Q where the degree of Q is nonnegative and P and Q have no zeros in common. The factorization theorem for polynomials leads at once to the conclusion that $M(\hat{\mathbb{C}})$ less the zero constant is the multiplicative group generated by the constants other than zero and the functions $\tau_a = z - a$, $z \in \hat{\mathbb{C}}$, $a \in \mathbb{C}$. From an additive point of view we see from the partial fraction

decomposition that $M(\hat{C})$ is the additive group generated by the functions $c\tau_0^n, c\tau_a^{-(n+1)}, a, c \in \mathbb{C}, n \in \mathbb{N}$.

EXERCISES

4.25. Show that the members of $M(\hat{C})$ having valence one are the members of $M(\hat{C})$ admitting a representation of the form $(az + b)/(cz + d)$ on C where $a, b, c, d \in \mathbb{C}$ and $ad - bc \neq 0$. Such functions are called *Möbius* (or *linear fractional*) transformations or *conformal automorphisms* of $\hat{C}$. Show that the Möbius transformations with the operation of composition constitute a group. Show that if a and b are distinct points of $\hat{C}$, there exists a Möbius transformation f satisfying $f(a) = 0, f(b) = \infty$.

Further properties of Möbius transformations will be developed in Ch. XV.

4.26. Show that D is a derivation on $M(\hat{C})$ which takes the constants into the zero constant if and only if there exists $g \in M(\hat{C})$ such that $D(f) = f'g, f \in M(\hat{C})$.

4.27. Show that the members of $M(C)$ of bounded valence are the restrictions to C of the nonconstant members of $M(\hat{C})$. Show that the valence of such a member of $M(C)$ has a constant value save at one point. What is the value of the valence at the exceptional point?

4.28. Show that the members of $M(\hat{C})$ having constant valence two are those representable in the form $\alpha \circ \tau_0^2 \circ \beta$ where α and β are Möbius transformations. Hint: Given that $f \in M(\hat{C})$ has valence two, consider its set of poles and proceed according as it has one or two members. The notation τ_0 is that of the preceding text.

4.29. $s = \frac{1}{2}(z + 1/z), z \in \hat{C}$, has constant valence two. Show that $s(z) = s(1/z)$ and that $s^{-1}([-1, 1]) = C(0; 1)$. Let P be a polynomial of nonnegative degree n and let

$$M = \max\{|P(x)|: -1 \leq x \leq 1\}.$$

Show that $|P \circ s(z)| \leq M|z|^n, 1 \leq |z| < +\infty$, with the aid of the following *boundary maximum principle* applied to $z^{-n}P \circ s(z), |z| > 1$:

Let O be an open proper nonempty subset of $\hat{C}$. Let $f: \bar{O} \rightarrow \mathbb{C}$ be continuous on $\bar{O}$ and analytic in O . Let $g = f|_{\text{fr } O}$. Then

$$\max_{\bar{O}} |f| = \max_{\text{fr } O} |f|.$$

[Proof: If $|f(a)| = \max_{\bar{O}} |f|$ for some $a \in O$, then by the maximum principle for analytic functions (cf. the sentences following (5.1) of Ch. IV) f is constant on the component of O containing a and hence $|f(a)| \leq \max_{\text{fr } O} |f|$. The asserted boundary maximum principle follows readily.]

Remark: The maximum principle for analytic functions remains in effect when the domain is a region in $\hat{C}$. The strong inequality in " $1 \leq |z| < +\infty$ " is there solely because $M|z|^n$ is not defined when $M = 0, z = \infty$. To be sure the question is trifling.

Show that for each positive integer n there exists a unique polynomial of degree n , say P , such that

$$z^n + \frac{1}{z^n} = P \circ s(z), \quad z \neq 0, \infty.$$

This is to be done without any formal computation. Hint: Consider

$$f = \left\{ \left(s(z), z^n + \frac{1}{z^n} \right) : z \in \hat{\mathbb{C}} \right\}$$

and show that $f \in \mathbf{M}(\hat{\mathbb{C}})$ and has a unique pole which is at ∞ and has order n .

4.30. Given $f: [-1, 1] \rightarrow \mathbb{C}$, a necessary and sufficient condition that f be the restriction to $[-1, 1]$ of a function g analytic at each point of $[-1, 1]$ is that, μ_n being the infimum of the set of elements

$$\sup \{ |f(x) - P_n(x)| : -1 \leq x \leq 1 \},$$

where P_n is a polynomial of at most degree n , we have

$$\limsup_{n \rightarrow \infty} (n+1)^{-1} \log \mu_{n+1} < 0.$$

(S. Bernstein). [$\log(+\infty)$ is defined as $+\infty$ and $\log(0)$ as $-\infty$.] Prove this theorem with the aid of the results of the preceding exercise. To show the condition necessary, consider $g \circ s(z)$ with domain $\{r < |z| < r^{-1}\}$, $0 < r < 1$, r sufficiently near one, and estimate the Laurent coefficients. Note that if (b_k) is the sequence of Laurent coefficients in question, $b_k = b_{-k}$ and conclude that $\sum_{k=-n}^n b_k z^k = P_n \circ s(z)$ where P_n is a polynomial of at most degree n . Estimate $|f(x) - P_n(x)|$, $-1 \leq x \leq 1$. To show the condition sufficient, note that there exists a sequence of polynomials (P_n) where P_n is at most of degree n and satisfies $|f(x) - P_n(x)| < r^n$, $-1 \leq x \leq 1$, where $0 < r < 1$, for n large. Apply the inequality of Ex. 4.29 to $P_{n+1} - P_n$ to conclude the uniform convergence of (P_n) in $s(\{r < |z| < r^{-1}\})$ and note that $f(x) = \lim_{n \rightarrow \infty} P_n(x)$, $-1 \leq x \leq 1$.

Limits of sequences of meromorphic functions. There remain to be considered among the more elementary questions of the theory of meromorphic functions limit problems and a useful and reasonable definition of *uniformity*. A convenient approach is to introduce following Carathéodory the so-called *chordal metric* for $\hat{\mathbb{C}}$. It induces the standard topology of $\hat{\mathbb{C}}$. We first introduce the *stereographic map* θ of the sphere S in $\mathbb{R}^3$ with center $(0, 0, \frac{1}{2})$ and radius $\frac{1}{2}$, that is, $\{x_1^2 + x_2^2 + x_3^2 - x_3 = 0\}$ onto $\hat{\mathbb{C}}$.† It is specified by the stipulation that it take $(0, 0, 1)$ into ∞ and each other point (x_1, x_2, x_3) of S into the image with respect to the canonical map σ of (y_1, y_2) , where $(y_1, y_2, 0)$ is the point of intersection of $\{x_3 = 0\}$ and the line passing through $(0, 0, 1)$ and (x_1, x_2, x_3) .

It is routine to verify that for $(x_1, x_2, x_3) \neq (0, 0, 1)$ we have

$$\theta(x_1, x_2, x_3) = \frac{x_1 + ix_2}{1 - x_3}.$$

It is easy to see that θ is a univalent continuous map of the compact set S onto $\hat{\mathbb{C}}$. Consequently φ , the inverse of θ , is a homeomorphism of $\hat{\mathbb{C}}$ onto S .

† Here $\mathbb{R}^3$ is $\{(x_1, x_2, x_3) : x_1, x_2, x_3 \in \mathbb{R}\}$.

Given $a, b \in \hat{\mathbb{C}}$, by the *chordal distance* between a and b , denoted $\chi(a, b)$, is meant the euclidean distance between $\varphi(a)$ and $\varphi(b)$, that is,

$$\left[\sum_1^3 (y_k - x_k)^2 \right]^{1/2}$$

where $\varphi(a) = (x_1, x_2, x_3)$ and $\varphi(b) = (y_1, y_2, y_3)$. That χ is a bona fide metric

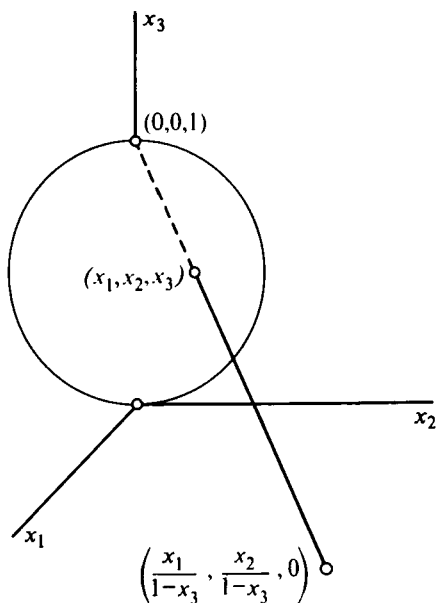


FIG. 6.1

on $\hat{\mathbb{C}}$ follows at once from the facts that $\chi(a, b)$ is just the euclidean distance between $\varphi(a)$ and $\varphi(b)$ and that φ is a univalent map. We call χ the *chordal metric* of $\hat{\mathbb{C}}$.

EXERCISE

4.31. Given $z \in \mathbb{C}$, show that

$$\varphi(z) = \left(\frac{\operatorname{Re} z}{1 + |z|^2}, \frac{\operatorname{Im} z}{1 + |z|^2}, \frac{|z|^2}{1 + |z|^2} \right)$$

and conclude that

$$\chi(a, \infty) = (1 + |a|^2)^{-1/2}$$

and

$$\chi(a, b) = |b - a|(1 + |a|^2)^{-1/2}(1 + |b|^2)^{-1/2}, \quad a, b \in \mathbb{C}.$$

Show that for $a, b \in \mathbb{C}$ we have the inequality

$$\chi(a, b) \leq |b - a|,$$

and when $a \in \mathbb{C}$ and $\chi(a, b) < \chi(a, \infty)$, the useful inequality

$$|b - a| \leq \frac{2}{\chi(a, \infty) - \chi(a, b)} \cdot \frac{\chi(a, b)}{\chi(a, \infty)}.$$

Hint: To establish the latter inequality proceed from

$$|b - a| \leq \chi(a, b)(1 + |a|^2)^{1/2}(1 + |a| + |b - a|),$$

and the inequality $(1 + |a|)^2 \leq 2(1 + |a|^2)$.

Show that for arbitrary $a, b \in \hat{\mathbb{C}}$ we have

$$\chi(1/a, 1/b) = \chi(a, b).$$

The map φ is an isometry of $\hat{\mathbb{C}}$ onto S . Hence φ is a homeomorphism of $\hat{\mathbb{C}}$ with the topology induced by χ onto S with its relative topology. Hence *the topology induced on $\hat{\mathbb{C}}$ by χ is the topology of $\hat{\mathbb{C}}$ introduced in §6, Ch. III.*

Uniformity questions will be referred to the *chordal metric*. Two basic questions call for immediate attention: (1) What is the nature of the uniform limit in the χ -sense of a sequence of meromorphic functions? (2) What is the relation between the notion of a uniform limit in the χ -sense and the usual one using the euclidean metric for sequences of functions taking values in $\mathbb{C}$?

(1) Let A be a region in $\hat{\mathbb{C}}$. Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of functions, each meromorphic on A , tending uniformly in the sense of χ in A to f . At the least, we know that f is a continuous map of A into $\hat{\mathbb{C}}$. Let $a \in A$ be such that $f(a) \in \mathbb{C}$. We fix positive r so that $\Delta(a; r) \subset A$ and

$$\chi[f(z), f(a)] < \frac{1}{2}\chi[f(a), \infty]$$

for $z \in \overline{\Delta(a; r)}$ and consider $m \in \mathbb{N}$ such that

$$\chi[f_n(z), f(z)] < \frac{1}{2}\chi[f(a), \infty]$$

for $z \in \overline{\Delta(a; r)}$, $n \geq m$. For such z and n we have $f(z) \in \mathbb{C}$, as well as

$$\chi[f(z), f_n(z)] < \frac{1}{2}\chi[f(a), \infty] < \chi[f(z), \infty],$$

and, using the second inequality of Ex. 4.31, we obtain

$$|f(z) - f_n(z)| \leq \frac{2}{\frac{1}{2}\chi[f(a), \infty] - \chi[f_n(z), f(z)]} \cdot \frac{\chi[f_n(z), f(z)]}{\frac{1}{2}\chi[f(a), \infty]},$$

whence we conclude that $|f(z) - f_n(z)|$ is arbitrarily small on $\overline{\Delta(a; r)}$ for n large. Hence f is analytic in $\Delta(a; r)$.

Suppose now that $f(a) = \infty$. From the last property of the chordal metric stated in Ex. 4.31 we see that $1/f$ is analytic at a and takes the value 0 there.

We now complete the discussion with the aid of the standard connectedness argument. Let E be the interior of $\{f(z) = \infty\}$. If $z \in A - E$, f is meromorphic at z and consequently $A - E$ is open. Since A is connected, either f is the constant ∞ or else f is meromorphic on A . We state this important result as a theorem.

Theorem 4.3: *Given A a region in $\hat{\mathbb{C}}$, $(f_n)_{n \in \mathbb{N}}$ a sequence of functions meromorphic on A tending uniformly in A to f in the sense of the chordal metric. Then either f is the constant ∞ or else f is meromorphic on A .*

(2) The relation between uniformity in the χ -sense and uniformity in the euclidean sense in the analytic case. From the first inequality of Ex. 4.31 we see that if f is the uniform limit in the sense of the euclidean metric of a sequence $(f_n)_{n \in \mathbb{N}}$ where each $f_n: X \rightarrow \mathbb{C}$, X being an arbitrary set, then f is also the χ -uniform limit of (f_n) . No demands are made on the nature of X . In the opposite sense we shall want to make use of analyticity in an essential way.

Suppose that the f_n of Theorem 4.3 are all analytic on A . Of course, the possibility that f is the constant ∞ must be taken into account (for example, $f_n(z) = n$). In this case we see that for each positive number p and each compact $C \subset A$, there exists $m \in \mathbb{N}$ such that $|f_n(z)| > p$, $z \in C$, $n \geq m$. If, however, f is meromorphic, it is not hard to see that f is even analytic on A . For if f is meromorphic on A and $a \in A$, there exists r positive such that $\overline{\Delta(a; r)} \subset A$ and $f(z) \neq \infty$, $z \in C(a; r)$. [$C(\infty; r) = \{|z| = r^{-1}\}$.] From the compactness of $C(a; r)$ and the local uniformity property established in Theorem 4.3, we see that (f_n) tends uniformly to f on $C(a; r)$ in the sense of the euclidean metric. Hence by the boundary maximum principle enunciated in Ex. 4.29, this section, we see that (f_n) is uniformly Cauchy on $\overline{\Delta(a; r)}$ and conclude that f is analytic at a . We see further that (f_n) tends uniformly to f in A in the sense of the euclidean metric.

EXERCISES

4.32. Prove that f of the preceding paragraph is analytic when it is meromorphic by appeal to Rouché's theorem.

4.33. Given $(f_n)_{n \in \mathbb{N}}$ where each f_n is a continuous map of A , an open subset of $\hat{\mathbb{C}}$ into $\hat{\mathbb{C}}$. Show that (f_n) tends uniformly to f in the sense of the chordal metric in A if

and only if for each $a \in A$ there exists r positive such that $\overline{\Delta(a; r)} \subset A$ and either $(f_{m+n})_{n \in \mathbb{N}}$ tends uniformly to f on $\overline{\Delta(a; r)}$ or $(1/f_{m+n})_{n \in \mathbb{N}}$ tends uniformly to $1/f$ on $\overline{\Delta(a; r)}$, in the euclidean sense, for some $m \in \mathbb{N}$, it being understood that the terms involved are meaningful in the sense of C. Show that the stated condition may be replaced by the following: For each $a \in A$ there exists r positive and a Möbius transformation M such that $(M \circ f_n)$ tends to $M \circ f$ uniformly in the euclidean sense on $\overline{\Delta(a; r)}$; the terms involved are to be meaningful in the sense of C. Can the continuity hypothesis on the f_n be dispensed with?

4.34. Let $A = \Delta(0; 1)$. $f_n(z) = 1/[z + (n+1)^{-1}]$, $g_n(z) = -(1/z)$. Show that although (f_n) and (g_n) have χ -uniform limits in A , $(f_n + g_n)$ does not. Let $h_n(z) = z - (n+1)^{-1}$. Show that $(f_n h_n)$ does not have a χ -uniform limit in A .

These examples show that algebraic operations on meromorphic functions may cause serious upsets for χ -uniformity.

4.35. For later applications the following situation is of interest. Each f_n is meromorphic on an open subset A in $\hat{\mathbb{C}}$. For each $a \in A$ for at most one n does f_n have a pole at a . Show that $\sum_0^\infty f_n$ is χ -uniformly convergent in A and has a meromorphic sum if and only if for each compact $C \subset A$ there exists $m \in \mathbb{N}$ such that f_{m+n} is analytic at each point of C , $n \in \mathbb{N}$, and $\sum_0^\infty f_{m+n} \upharpoonright C$ converges uniformly in the sense of the euclidean metric.

A theorem of Hurwitz. An important question concerning sequences of meromorphic functions is the relation of the sets $\{f_n(z) = b\}$ to $\{f(z) = b\}$ where $b \in \hat{\mathbb{C}}$ and $(f_n)_{n \in \mathbb{N}}$ is a sequence of functions, each meromorphic on a region A in $\hat{\mathbb{C}}$, which tends uniformly in the sense of the chordal metric in A to f . If f is the constant function with value b , nothing much can be said. Otherwise we can make a fairly precise statement which is the substance of the following theorem of A. Hurwitz.

Theorem 4.4: *Under the stated assumptions, f not being the constant b , if $f(a) = b$ and r is a positive number such that $\overline{\Delta(a; r)} \subset A$ and $f(z) \neq b$, $z \in \overline{\Delta(a; r)} - \{a\}$, then there exists $m \in \mathbb{N}$ such that the value of the valence of $f_n \upharpoonright \overline{\Delta(a; r)}$ at b is precisely $n(a; f)$ for $n \geq m$.*

PROOF: It is based on Rouché's theorem, as we might expect from the very statement. Let M denote a Möbius transformation satisfying $M(b) = 0$. We take s , $0 < s < r$, such that $|M \circ f(z)| < 1$, $z \in \overline{\Delta(a; s)}$, and introduce

$$\mu = \min\{|M \circ f(z)| : z \in \overline{C(a; s)}\}.$$

By the hypothesis on r we have $\mu > 0$. Since $(M \circ f_n)$ tends χ -uniformly to $M \circ f$ in A , when n is large, $M \circ f_n(z) \in \mathbb{C}$, $z \in \overline{\Delta(a; s)}$ and

$$|M \circ f_n(z) - M \circ f(z)| < \mu, \quad z \in \overline{C(a; s)}.$$

Hence by Rouché's theorem

$$M \circ f_n = M \circ f + (M \circ f_n - M \circ f)$$

has as many zeros in $\Delta(a; s)$, counted by multiplicity, as does $M \circ f$ when n is large, so that for n large the valence at b of $f_n | \Delta(a; s)$ is $n(a; f)$. For n large, $f_n(z) \neq b$, $z \in \Delta(a; r) - \Delta(a; s)$, a set which is compact and such that f does not take the value b on it. The theorem follows.

EXERCISES

4.36. Let O and g satisfy the conditions stated in Rouché's theorem with f the constant zero. Suppose that $g_n: \hat{O} \rightarrow \hat{\mathbb{C}}$ is continuous on $\hat{O}$ and is meromorphic in O , $n \in \mathbb{N}$, and that (g_n) tends χ -uniformly on $\hat{O}$ to g . Then for n sufficiently large $\nu_{g \circ f}(0) = \nu_{g_n \circ f}(0)$ and $\nu_{g \circ f}(\infty) = \nu_{g_n \circ f}(\infty)$.

What can be concluded about ν_{f_n} where $f_n \in \mathbf{M}(\hat{\mathbb{C}})$ and (f_n) tends χ -uniformly on $\hat{\mathbb{C}}$ to f ?

4.37. Let A and B be regions $\subset \hat{\mathbb{C}}$. Let $(f_n)_{n \in \mathbb{N}}$ be a sequence such that $f_n: A \rightarrow B$ and is meromorphic on A , $n \in \mathbb{N}$, and (f_n) tends χ -uniformly to f in A . Then either $f(A) \subset B$ or f is constant.

4.38. Let $(f_n)_{n \in \mathbb{N}}$ denote a sequence of functions meromorphic on a region A in $\hat{\mathbb{C}}$ tending χ -uniformly to f in A . Suppose that f is not constant and that $\alpha < \nu_f(w)$. Then $\alpha < \nu_{f_n}(w)$ for n sufficiently large. Conclude that if

$$\sup \{ \nu_{f_n}(w) : w \in \hat{\mathbb{C}} \} \leq \beta < +\infty$$

for all n , then either $\nu_f(w) \leq \beta$, $w \in \hat{\mathbb{C}}$, or f is constant. In particular, if each f_n is univalent, then f is univalent or constant.

4.39. We have seen that algebraic operations need not combine well with the taking of χ -uniform limits. The object of the present exercise is to remedy the situation and to account for the breakdown by "coalescence." Specifically, we suppose that (f_n) tends χ -uniformly to f and (g_n) χ -uniformly to g in the region $A \subset \hat{\mathbb{C}}$, f_n, f, g_n, g being meromorphic on A . Then $(f_n + g_n)$ tends χ -uniformly to $(f + g)$ in

$$A - \{f(z) = \infty\} \cap \{g(z) = \infty\},$$

and $(f_n g_n)$ tends χ -uniformly to $f g$ in

$$A - \{[\{f(z) = \infty\} \cap \{g(z) = 0\}] \cup [\{f(z) = 0\} \cap \{g(z) = \infty\}]\}.$$

4.40. Let A be a region in $\hat{\mathbb{C}}$ and X a compact metric space. Develop a theory of $F: A \times X \rightarrow \hat{\mathbb{C}}$ continuous on $A \times X$ and such that for each $p \in X$ the function $f_p(z) = F(z, p)$, with domain A , is meromorphic on A (cf. §6, Ch. V). In particular, investigate (1) the function $f_p'(z)$, with domain $A \times X$, and (2) the properties of ν_{f_p} for p near $q \in X$, formulating and establishing appropriate theorems.